

Why Only 3D for 3D?

2D-Guided Supervision for 3D Medical Vision-Language Models

Qilong Zhao · Ziyuan Qin · Liang Zhao



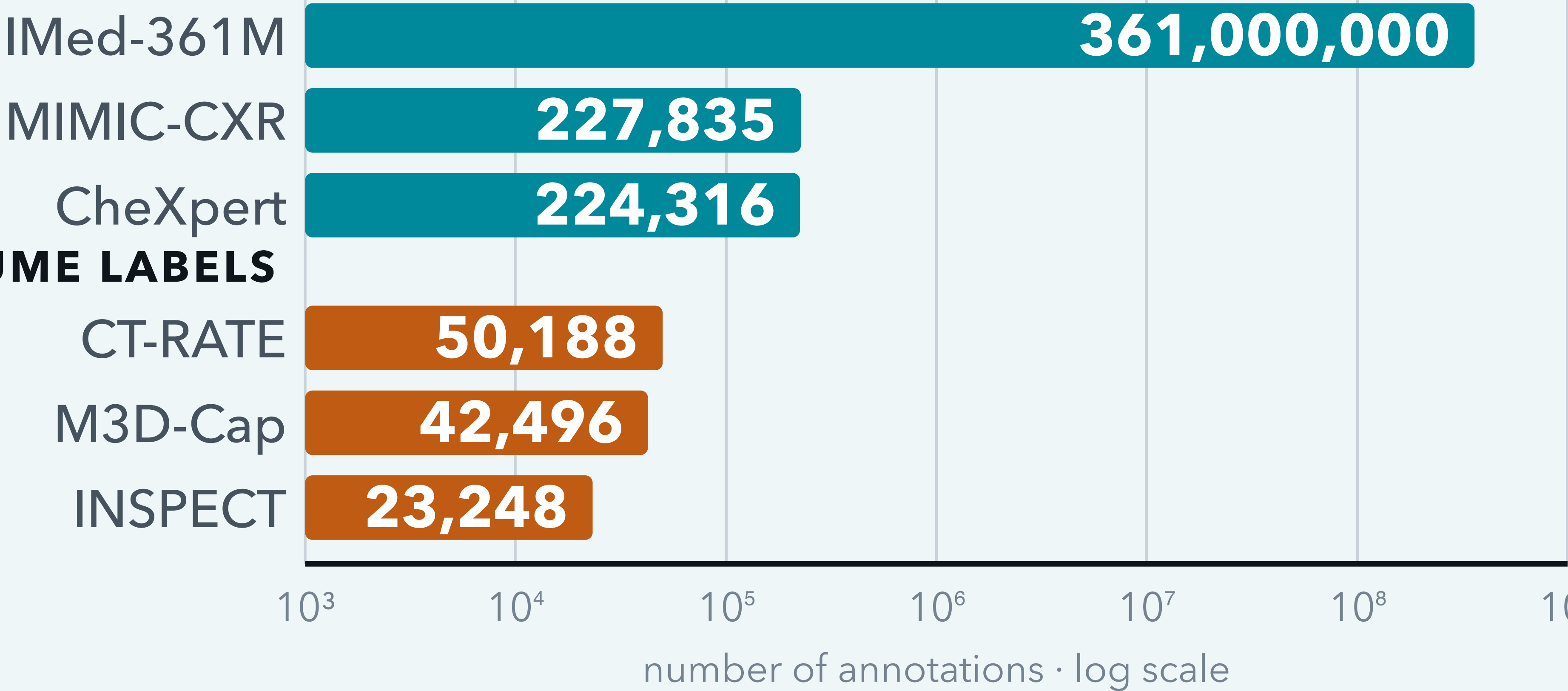
1 3D medical AI is starved for labels

Reading CT and MRI **volumes** takes expert-labelled volumes – and those barely exist. Labels on single **slices** are everywhere.

8,500×

more labels in the largest **2D** collection than in the largest **3D** one

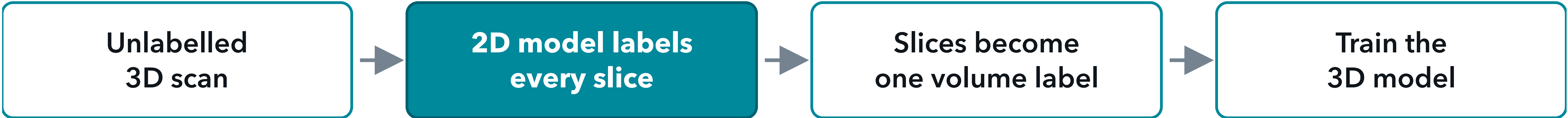
2D · SLICE LABELS



3D · VOLUME LABELS

2 So let a 2D model do the labelling

TRAINING



AT INFERENCE



The 2D model only makes labels – it is gone at inference.

3 What we found

Machine labels beat expert ones when labels are scarce

Reports written by a 2D model, CT. The two converge once labels are plentiful.

Expert reports Machine reports

Clinical F1, micro · higher is better

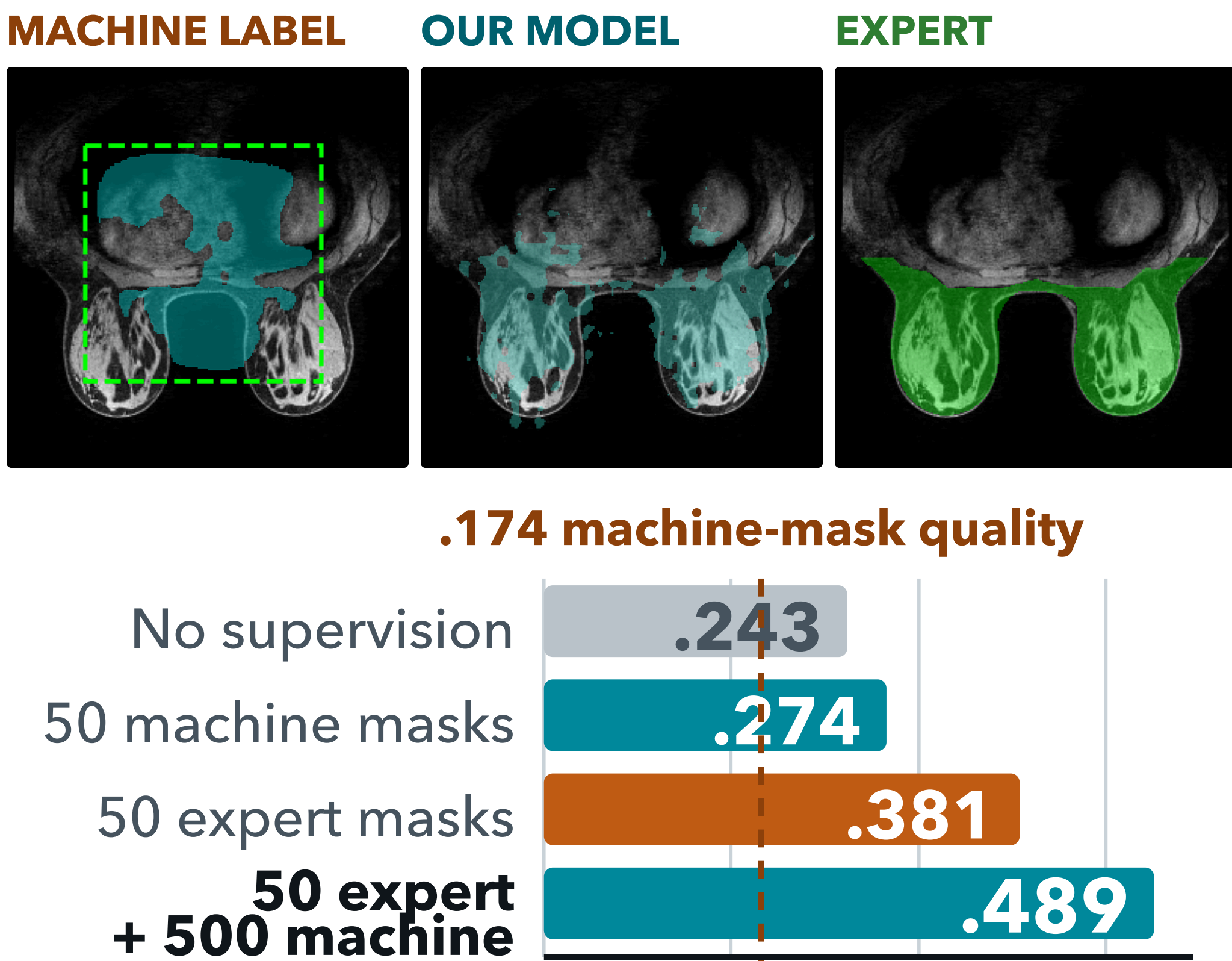
+33% when labels are scarce



Even poor labels still teach

Machine masks score only 0.174 themselves – yet they lift the 3D model. Duke breast MRI.

No supervision Expert masks Machine masks added
Dice · higher is better



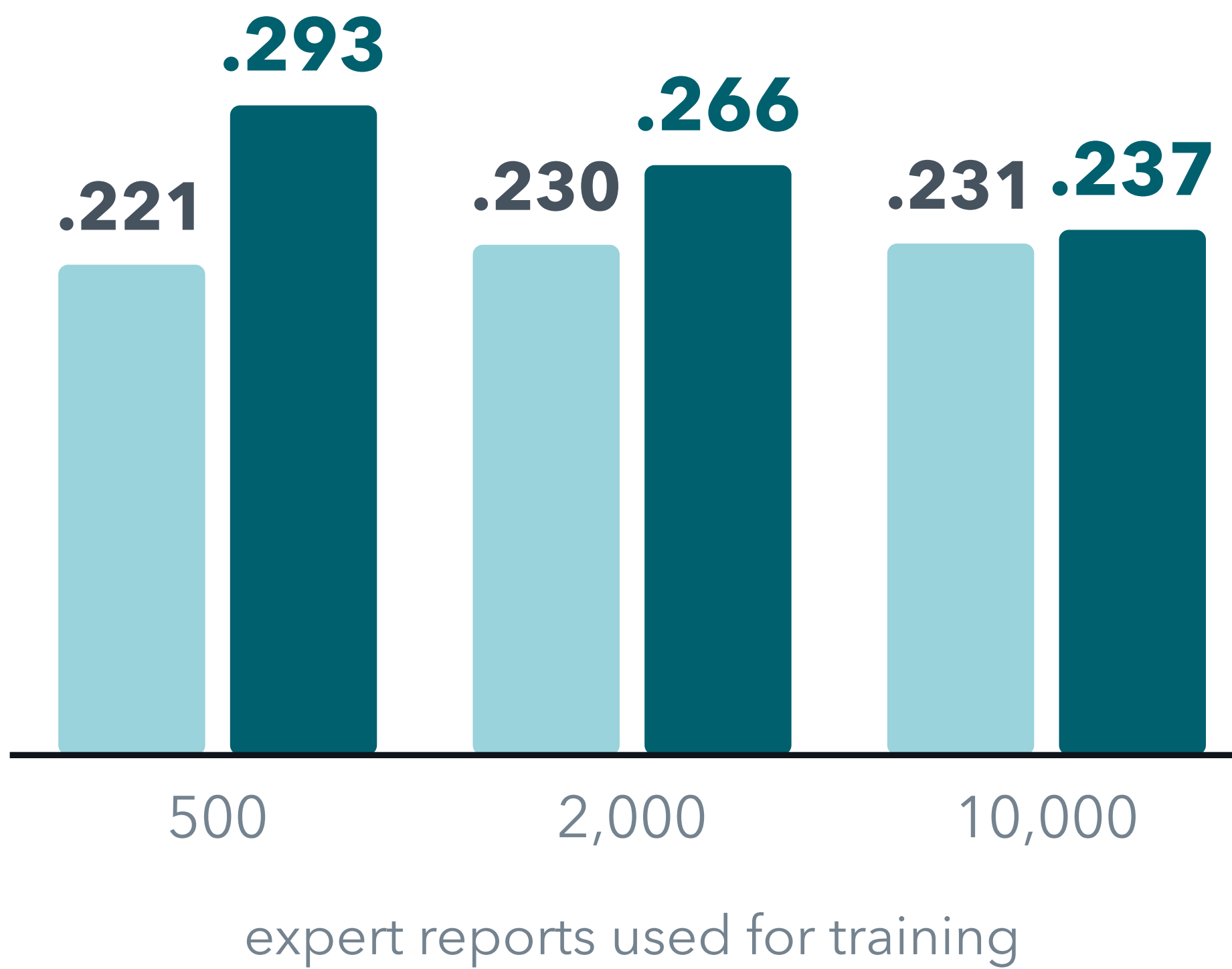
A better 2D teacher lifts the 3D student

Swap in a stronger 2D model and regenerate the labels. Nothing about the 3D model changes.

Qwen2-VL teacher Qwen3-VL teacher

Clinical F1, micro · higher is better

Qwen2-VL → Qwen3-VL



4 It stops the model making things up

WHAT IS IN THE SCAN

A 38 mm mass at the tail of the pancreas, masses in the liver, and small cysts in the left kidney.

To recover: mass, cyst

100 EXPERT REPORTS

F1 0.500

“A large, heterogeneous, enhancing **occipital soft tissue mass**, with **intracranial extension**, measuring 11.5 × 9.8 × 6.7 cm.”
Wrong organ – a fabricated head mass.

+ 200 MACHINE REPORTS

F1 1.000

“A 12 mm enlarging **cystic lesion** in the right adrenal gland, with no **enhancing solid component**.”
Right region, both findings recovered.

Abundant 2D expertise can supervise scarce 3D learning.

Gains are largest where expert labels are fewest. Report generation on M3D-Cap (CT), segmentation on Duke Breast MRI (50 train / 50 test). Research use only – not validated for clinical deployment.

alexqilong@gmail.com · liang.zhao@emory.edu



Code